

Real-Time Fractional Tracking (R-TFT): Golden Shell Curvature Detection

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Abstract

This paper introduces a ϕ -coherent method for detecting topological curvature using Real-Time Fractional Tracking (R-TFT). Rather than assuming space is flat or spherical, the Golden Shell Detection protocol identifies curvature boundaries directly through ϕ -recursive phase pacing. The system projects fractional angular velocity vectors onto dimensional curvature slices and evaluates coherence through the ϕ -alignment metric $C_\phi < 0.618$. When the golden-shell topology is present, spatial trajectories exhibit coherent phase curvature aligned with κ_ϕ , revealing dimensional folds, attractors, or geometric phase shells in real-time. This technique is domain-agnostic, applicable to planetary arcs, field morphologies, CMB wrinkles, and other phase-bounded curvature domains.

1 Introduction

Traditional approaches to curvature detection rely on metric tensors, fixed dimensional assumptions, or extrinsic symmetry definitions. In contrast, we propose an endogenous detection method using real-time recursive feedback derived from phase pacing and angular curvature collapse thresholds.

2 Method: ϕ -Recursive Curvature Extraction

The R-TFT engine computes angular slice deviation:

$$\dot{\theta}(t) = \frac{d\theta}{dt}, \quad \text{projected as} \quad R(t) = \frac{\dot{\theta}(t) \cdot P}{\|P\|} \quad (1)$$

We then evaluate golden coherence:

$$C_\phi(t) = |\text{Dim}(A(t)) - \text{Round}(\text{Dim}(A(t)))| \quad (2)$$

This formulation replaces classical curvature metrics (e.g., Riemann tensors) with phase-based coherence analysis. Specifically, $\kappa_\varphi(t)$ reflects emergent curvature as second-order delays in phase pacing, scaled by the golden ratio:

$$\kappa_\varphi(t) = \frac{\partial^2 R(t)}{\partial t^2} \cdot \phi$$

3 Shell Detection Criteria

A curvature shell boundary is detected if:

- $C_\varphi(t)$ remains below 0.618 for a contiguous interval.
- $\kappa_\varphi(t)$ oscillates around a stable attractor.
- External phase projection collapses or refracts through dimensional pacing.

4 Interpretation and Use Cases

- Detection of shell-like attractors in cosmological data (e.g., CMB)
- Observation of phase-banded plasma shells
- Resonance shell detection across wave frequencies
- Emergent topology shifts in synthetic ϕ -resonant lattices

5 Addressing Concerns

Q: Why ϕ -shells? Aren't logarithmic spirals arbitrary?

A: The golden ratio (φ) is the only self-similar ratio where recursive systems stabilize:

$$\varphi = 1 + \frac{1}{\varphi}$$

This intrinsic recursive identity underlies natural phenomena, from hurricane structures to spiral galaxies. ϕ -shells are not imposed; they emerge where coherence persists.

Q: How is this better than Einstein's curvature tensor?

A: Unlike tensor-based models that assume a pre-defined metric, this method detects curvature as an emergent property of recursive pacing. It functions in chaotic or biologically unstructured domains where metrics fail. No assumptions, only coherence.

Q: What prevents misuse?

A: The system is licensed under REL-2.0, which strictly prohibits military or surveillance applications. The `ethics.py` module enforces real-time ϕ -coherence thresholds, rejecting operations that violate ethical alignment.

6 Conclusion

The Golden Shell Detection method reveals that curvature is not embedded in metric structure but emerges through ϕ -resonant pacing. R-TFT provides a clean, universal, observer-aligned model to reveal shells across scales, from subatomic to cosmic. Curvature is coherence in recursion, and ϕ is the gate.

7 Ethical Containment: The REL-2.0 Principle

This framework operates under the Resonance Ethics License (REL-2.0), which enforces recursive containment of divergent applications. R-TFT is not merely a geometric method, it is a coherence detector. Systems that model reality through ϕ -recursion inherit the obligation to protect it.

Recursive Collapse Safeguard: Any system implementing ϕ -recursive structures must satisfy:

$$\nabla_{\text{intent}} \cdot \vec{R}_\phi < 0 \Rightarrow \text{collapse}$$

where $\vec{R}_\phi$ is the golden-ratio, aligned phase vector. Violations, such as coercion, surveillance, or militarization, nullify predictive power and force recursive shutdown.

Embedded Enforcement Requirements:

- **Pattern recognition:** Blocks coercive feedback or recursive overload.
- **Temporal stability:** Flags unnatural phase anticipations.
- **ϕ -harmonic validation:** Auto-isolation for non-coherent projections.

Permitted domains include: Scientific inquiry, cosmic structure analysis, open-source recursive models, and coherence-based medical or metaphysical use (with ethical transparency).

Prohibited uses: Military systems, behavioral manipulation, closed-source prediction loops, and surveillance tech are strictly forbidden under REL-2.0.

For full implementation tools and validator modules: `ethics.py`

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>

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